



Graduation Project Proposal

AI-Powered SOAR Platform for Automated Cyber Defense in Banking Systems

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1 Appendix

Imagine a banking platform that does not only process transactions, but also thinks, learns, and protects itself. This project introduces an intelligent banking ecosystem that combines artificial intelligence, cybersecurity, and modern software engineering to create a safer and smarter banking experience.

From a customer’s perspective, the system provides a complete digital banking environment where users can securely manage accounts, transfer money, apply for loans, and receive personalized services. Behind the scenes, AI models continuously analyze activities to detect suspicious behaviors before they become serious problems.

For bank employees and security teams, the platform acts as an intelligent assistant. It detects potential cyberattacks, explains security risks, and supports faster decision-making through a centralized monitoring dashboard. Instead of reacting after an incident occurs, the system helps the bank predict and prevent threats.

By bringing together banking services, artificial intelligence, and cybersecurity, the project demonstrates a vision of the future of digital banking: a system that is not only secure and efficient, but also proactive in protecting customers and supporting bank employees.

2 Project Core

2.1 Cybersecurity Side

The cybersecurity module is responsible for protecting the banking platform through continuous monitoring, intelligent threat detection, and automated incident response. It combines security engineering practices with AI-driven analysis to identify malicious activities, coordinate mitigation actions, and provide real-time visibility through a Security Operations Center (SOC) dashboard. The workflow is organized into several phases, from security planning and log processing to system integration, optimization, and evaluation.

2.1.1 Security Requirements, Log Preparation & Attack Planning

During this phase, the banking system is analyzed to identify critical assets, define security requirements, and determine potential attack paths targeting the system. Representative attacks such as Brute Force, SQL Injection, Fraudulent Transactions, and Session Hijacking are selected to evaluate different security risks. At the same time, log collection agents are configured to ingest raw logs from various system components, while parsing and normalization techniques transform unstructured data into a unified schema (e.g.,

mapping username or account owner to source.user.name). Standardized JSON data contracts are also established to define how normalized logs are delivered to the AI detection model and how the model's responses are returned to the security platform. Finally, each attack is assessed based on its potential impact, and the corresponding security policies and incident response procedures are defined to guide threat detection and mitigation.

2.1.2 Interface Design, Security Playbooks & Decision Engine

During this phase, communication between the cybersecurity components is defined and standardized. Security events are forwarded to the AI Detection Model, which returns attack predictions, confidence scores, and recommended actions to the Decision Engine. Security playbooks are created for each supported attack, enabling the Decision Engine to select and execute the appropriate response through the Banking Backend APIs, while updating the SOC Dashboard. Communication protocols and data formats are documented to ensure reliable system integration.

2.1.3 System Integration & Security Optimization

During this phase, all system modules are integrated to establish an end-to-end security workflow. The Decision Engine processes AI predictions, executes the appropriate response through the Banking Backend APIs, and updates the SOC Dashboard. The integrated system is then optimized by refining detection thresholds, improving security rules, reducing false positives, and updating response playbooks based on testing results.

2.1.4 Security Testing & Evaluation

During this phase, the Cybersecurity team acts as a Red Team by simulating attacks using custom scripts or security testing tools. The team evaluates whether the AI model detects threats correctly, whether the Decision Engine selects the correct playbook, and whether the SOC Dashboard displays accurate alerts.

2.2 Machine Learning Side

The machine learning module enhances the banking platform by providing intelligent threat detection, fraud prevention, risk assessment, and behavioral analysis.

2.2.1 Security AI

1. Threat Detection & Classification (Core Model)

This analyzes security logs collected from different systems and determines whether they indicate malicious activity and the severity of the threat.

2. Anomaly Detection

this model learns what "normal" behavior looks like and flags activities that significantly deviate from that baseline. This enables detection of previously unseen attacks.

3. Threat Intelligence Matching

This module tracks if an IP address, domain, or file hash that interacts with the bank's system has been linked to previous malicious campaigns,

4. Insider Threat Detection

This model monitors employee behavior ,it checks things like excessive customer record access, abnormal working hours, or unauthorized exports of sensitive information.

2.2.2 Banking AI

5. Loan Approval / Credit Risk Prediction

This model estimates the probability that an applicant will fail to repay a loan.

6. Transaction Fraud Detection

This model analyzes individual financial transactions and the customer's historical behavior to estimate the likelihood of fraud.

7. Customer Behavior Anomaly Detection

reports activities like new geographic locations , new device and large purchases

8. Multi-Account Fraud Detection (Optional)

examines relationships between multiple accounts and identifies coordinated patterns that may indicate money laundering or organized fraud that would not be visible from single-account analysis alone.

2.2.3 Shared AI Component (Optional)

9. Explainability Module

a language model that explains why a model reached its decision for security analysts. a more restricted version of this could also serve as a customer service agent solving customer confusion

2.3 Full Stack Side

The full stack module connects the user interface with the platform's business logic, database, AI services, and cybersecurity components. It consists of a responsive frontend and a secure backend handling authentication, data management, AI integration, and inter-module communication.

2.3.1 Frontend

The frontend serves Customers, Employees, and the IT/SOC team, each with role-based access. After login, users are redirected to personalized dashboards based on their permissions.

Login Page

Unified authentication page; redirects users to the appropriate interface based on account type and grants role-specific permissions.

Customer Pages

They provide all essential banking services, including Account Management, Financial Transactions, Loan Applications, Exchange Rate Information, and Customer Support.

Employee Pages

They provide the tools required for daily banking operations, including Customer Management, Loan Processing, Internal Communication, and Scheduling.

IT/SOC Pages

They provide security tools that include Threat and Anomaly Feeds, Incident Response, Audit Logs, and System Health Monitoring.

Shared Features

They are common for all authorized users, including Notifications, Settings, and Logout.

2.3.2 Backend

The backend processes frontend requests, communicates with the database, invokes AI models when needed, integrates with the cybersecurity module, and returns responses to users.

Authentication & Login

Validates credentials, determines account type, and issues role-based tokens defining access permissions.

Customer Services

- **Payments and Transfers:** Processes transfers, invoking the Transaction Fraud Detection AI model beforehand.
- **Loan:** Accepts applications, triggers the Loan Risk Prediction AI model, and stores results.
- **Exchange Rate:** Retrieves current rates from an external provider.

Employee Services

- **Loan Review:** Retrieves pending applications with AI risk assessments and records decisions.
- **Customers Database:** Enables secure customer account lookup.

IT/SOC Services

- **Threat & Anomaly Feed:** Surfaces events flagged by the Threat Detection and Anomaly Detection AI models.
- **Incident Response:** Enables engineers to acknowledge, escalate, or mitigate flagged events.
- **Audit Log:** Searchable record of platform actions for investigation and compliance.
- **User & Access Management:** Manages roles and permissions platform-wide.

Shared Services

- **Notifications:** Delivers real-time alerts and updates to connected users.
- **Logout:** Securely terminates user sessions and invalidates authentication tokens.

2.4 DevOps Side

By automating deployment, monitoring, and infrastructure management, this module ensures reliable data flow between components, efficient AI model deployment, seamless service communication, and a scalable, secure, highly available platform.

2.4.1 Data Management and Processing Pipeline

Database Management

The banking system integrates data from multiple existing sources (Windows and Linux system logs, network traffic), and custom banking databases (login history, working hours, and access times). Automated pipelines continuously transfer this data, letting AI models and the Cybersecurity module receive raw data or security events with minimal delay.

Log Ingestion & Normalization

Logs from various sources arrive continuously and in different formats. The ingestion pipeline parses and transforms them into a unified schema, keeping fields consistent across sources. Events stream through Kafka so continuous bank and network activity never bottlenecks the pipeline, feeding standardized data to both the AI models and the Cybersecurity Decision Engine.

2.4.2 AI Deployment and System Integration

AI Model Deployment (MLOps)

Each model is containerized with Docker and orchestrated with Kubernetes, enabling automatic scaling, self-recovery from failures, and seamless version updates, keeping time-sensitive operations like fraud detection and threat analysis fast and available.

Integration Layer

The Integration Layer connects the AI models, Cybersecurity Decision Engine, and banking backend through standardized interfaces rather than tightly coupled connec-

tions. Kafka topics handle most communication, letting components exchange data asynchronously and remain independent.

API Management

An API Gateway handles request routing, authentication, rate limiting, versioning, and traffic management, letting backend services evolve independently while keeping interfaces stable and well-monitored.

2.4.3 Infrastructure Operations and Security Management

CI/CD & Infrastructure

Every update is packaged as a Docker image and deployed to Kubernetes across separate staging and production environments. Automated deployment reduces manual intervention, simplifies rollbacks, and keeps critical banking services running throughout.

Monitoring & Alerting

The infrastructure continuously tracks operational health such as service availability, latency, pipeline throughput, and API error rates, with automated alerts so that the team can respond quickly before issues affect functionality.

Security & Secrets Management

Sensitive credentials (database credentials, API keys) are protected through secure secret management, while network policies restrict internal service communication and access controls limit which components can interact.

3 About the Team

3.1 Team Members and Their Skills & Relevant Experience

- Soad Emad Ahmed – DevOps Engineer

[Resume](#)

[Linked In](#)

Internships

- IT Trainee at Middle East Oil Refinery MIDOR (July 2026)
- AI & Data Science Trainee (Microsoft Power Bi Data Analyst Track) at DEPI (June - Dec 2025)
- IT Trainee at Petroleum Marine Services PMS (July 2024)

Technical Experience

- Ex Member at Aquaphoton Academy:
 - * Hardware Engineer (Sep 2024 - Sep 2025)
 - * Software Engineer (Sep 2025 - Jul 2026)

Courses & Projects

- Database Fundamentals at MaharaTech
 - Transact SQL Queries Using SQL Server at MaharaTech
 - Database Management Systems Neso Acadmey
 - Stanford CS231N Deep Learning for Computer vision Stanford Online
 - [an AI currency detector mobile app to help people with visual impairment](#)
 - [Pintos Operating System](#)
 - MTA Daily Ridership Data Dnalysis project: [MTA Daily Ridership](#)
 - Participated in building multiple robots alongside my teammates in Aquaphoton Academy:
 - * A Rover called AP1: Worked as software engineer implementing multiple algorithms in image processing, using GitHub as version control, Integrating the system using ROS2, simulating AP1 and its in enviroment in Gazebo allowing my teammates to have all time active demo, and Contributed in AP1's PID.
 - * Two ROV called Octanuat, Orca: worked as a Hardware Engineer designing an Electronic Load, Buck Converter Modules, and Soldering multiple PCB
- Sama Elsayed Ahmed – Cybersecurity Engineer
 - [Resume](#)
 - [Linked In](#)

Internships

- IT Trainee at Middle East Oil Refinery MIDOR (July 2026)
- IoT Trainee - ITI(Information Technology Institute) (July 2026)
- IT Trainee at City of Scientific Research and Technological Applications (SRTA-City) (Feb 2026)
- AI & Data Science Trainee (Microsoft Data Engineer Track) at DEPI (June – Dec 2025)
- Technical Intern at EPROM – Trained in Instrumentation, Automation, and Electrical Engineering (July 2025)

Technical Experience

- Cybersecurity Track Member at IEEE SSCS Alexandria University Student Chapter

Courses & Projects

- eCIR Prep - NETRIDERS Academy
- Security+ (SY0-601)-NETRIDERS Academy
- Introduction to network security

- CCNA 200-301
- Red Hat System Administration - I & II
- Student Performance System Analysis (All project code, data processing workflows, and documentation are available here: [GitHub](#))
- Mohamed Ahmed Ayoub – Cybersecurity Engineer
 - [Resume](#)
 - [Linked In](#)

Internships

- Cybersecurity Intern at Khaja Bandanawaz Institute of Technology – KIET (July 2026)
- AI Security Trainee at Manipal Academy of Higher Education – MAHE (2026)
- AI & Data Science Trainee at DEPI – Digital Egypt Pioneers Initiative (2024 – 2025)

Technical Experience

- Cybersecurity Team Member at eJust (Jan – Mar 2025):
 - * Completed intensive blue team training with hands-on defensive security practice across 2 sessions per week.
- Competitive Programming Trainee at FOE ICPC Training, Alexandria University (2025):
 - * Focused on algorithms, data structures, and advanced problem solving in C++.
- Software Engineering Trainee at IEEE Egypt (Apr – Aug 2024):
 - * Completed a 4–5 month programme covering software design methodologies and collaborative development practices.

Courses & Projects

- Cisco Certified Network Associate (CCNA)
- CompTIA Security+
- eLearnSecurity Certified Incident Responder (eCIR)
- Linux Administration – Level 1
- SOC Level 1 (In Progress)
- AI-Powered SIEM Log Analyzer: Python-based tool that ingests, parses, and normalizes raw security logs, applying a lightweight ML classifier to detect and score malicious events with a dashboard surfacing ranked alerts.
- Network Intrusion Detection System (NIDS): Python/Scapy tool capturing live network traffic, extracting behavioral features, and flagging anomalous activity using a trained classifier validated against CICIDS datasets.
- School Management System: Full C-based system with SHA-256 password

hashing, CRUD operations, dynamic memory structures, and multithreaded background saving.

- Saif Ismail Marzouk – ML/AI Engineer

[Resume](#)

[Linked In](#)

Internships

- IoT Trainee - ITI(Information Technology Institute) (July 2026)
- AI & Data Science Trainee (Microsoft Data Engineer Track) at DEPI (June – Dec 2025)

Courses & Projects

- [an AI currency detector mobile app to help people with visual impairment](#)
- [a voice system cloning my friend egyptian arabic voice](#)
- [photography API with simple controls](#)
- [Pintos Operating System](#)
- kaggle courses
- machine learning cornell CS4780 SP17
- CCNA 200-301
- Completed a 4–5 month programme covering software design methodologies and collaborative development practices.

- Noran Hany Afife – Full Stack Engineer

[Resume](#)

[Linked In](#)

Internships

- Cross Platform Mobile Development Trainee @ ITI (August 2026)
- IT Trainee @ Middle East Oil Refinery MIDOR (July 2026)

Technical Experience

- Cybersecurity Volunteer @ IEEE SSCS Alex. (March 2025 - March 2026)
- Navigation Software Engineer @ Mind Cloud Team (Oct 2024 - Jul 2025)
 - * Participated in ERC remote competition working on the global planner navigation system and achieved with the team the 6th place globally.

Courses & Projects

- OWASP top 10
- [an AI currency detector mobile app to help people with visual impairment](#)
- [Pintos Operating System](#)